

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 11 · Applications of Differentiation

With Answers

Questions with full worked solutions

- 52 pages
- Past papers up to January 2026

PURE 2 | TOPIC 11

Applications of Differentiation

Your classified route

20

QUESTIONS

175

MARKS

4

FAMILIES

5

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 11 | FAMILY 01

Gradient/Rate

WHAT THIS FAMILY ASKS

Recognise the gradient/rate demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**2 classified questions**

19 total marks

T11

11_GRAD

11_GRAD_CORE

CHAPTER
Applications of Differentiation

FAMILY
Gradient/rate

VARIANT
core family move

7 marks

WMA12/01 OCTOBER 2019
Question 10

Past-paper question

WMA12-01_2019_Oct_Q10

10. The curve C has equation

$$y = ax^3 - 3x^2 + 3x + b$$

where a and b are constants.

Given that

- the point $(2, 5)$ lies on C
- the gradient of the curve at $(2, 5)$ is 7

(a) find the value of a and the value of b .

(4)

(b) Prove that C has no turning points.

(3)

T11 Applications of
Differentiation

11_GRAD Gradient/rate

11_GRAD_CORE core family
move

WORKED SOLUTION

Question 10

WMA12/01 October 2019

WMA12-01_2019_Oct_Q10

1

Use the point on the curveSubstitute $(2, 5)$ into $y = ax^3 - 3x^2 + 3x + b$: $5 = 8a - 12 + 6 + b$, so $8a + b = 11$.

2

Differentiate

$$\frac{dy}{dx} = 3ax^2 - 6x + 3.$$

3

Use the given gradientAt $x = 2$, the gradient is 7, so $12a - 12 + 3 = 7$. Hence $12a = 16$ and $a = \frac{4}{3}$.

4

Find b

Using $8a + b = 11$, $b = 11 - 8\left(\frac{4}{3}\right) = \frac{1}{3}$.

5

Check for turning points

With $a = \frac{4}{3}$, $\frac{dy}{dx} = 4x^2 - 6x + 3$.

6

Use the discriminant

For $4x^2 - 6x + 3 = 0$, the discriminant is $(-6)^2 - 4(4)(3) = 36 - 48 = -12 < 0$.

7

Conclude

The derivative is never zero, so the curve has no turning points.

FINAL ANSWER(a) $a = \frac{4}{3}$, $b = \frac{1}{3}$. (b) **Proven: C has no turning points.**

WMA12-01_2019_Oct_Q10

T11

11_GRAD

11_GRAD_CORE

CHAPTER
Applications of Differentiation

FAMILY
Gradient/rate

VARIANT
core family move

12 marks

WMA12/01 OCTOBER 2023

Question 9

Past-paper question

WMA12-01_2023_Oct_Q09

9.

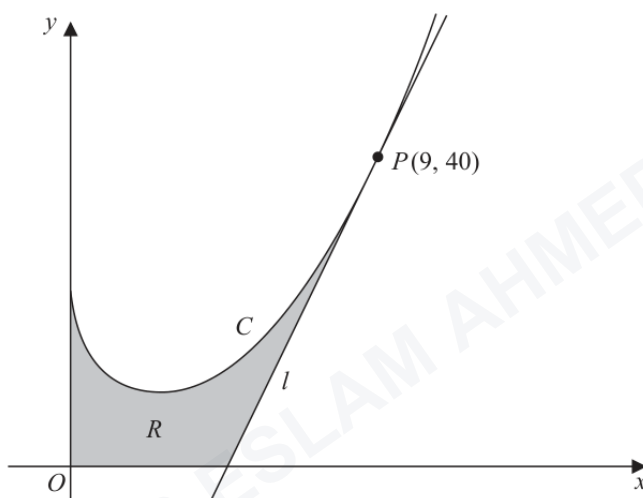


Figure 3

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Figure 3 shows a sketch of part of the curve C with equation

$$y = \frac{2}{3}x^2 - 9\sqrt{x} + 13 \quad x \geq 0$$

- (a) Find, using calculus, the range of values of x for which y is increasing.

(4)

The point P lies on C and has coordinates $(9, 40)$.

The line l is the tangent to C at the point P .

The finite region R , shown shaded in Figure 3, is bounded by the curve C , the line l , the x -axis and the y -axis.

- (b) Find, using calculus, the exact area of R .

(8)

T11 Applications of
Differentiation

11_GRAD Gradient/rate

11_GRAD_CORE core family
move

WORKED SOLUTION

Question 9

WMA12/01 October 2023

WMA12-01_2023_Oct_Q09

1

DifferentiateFor $y = \frac{2}{3}x^2 - 9\sqrt{x} + 13$, $\frac{dy}{dx} = \frac{4}{3}x - \frac{9}{2\sqrt{x}}$.

2

Set up increasing condition y is increasing when $\frac{dy}{dx} > 0$.

3

Multiply by a positive quantityFor $x > 0$, multiply by $6\sqrt{x}$: $8x\sqrt{x} - 27 > 0$.

4

Solve the inequality $x^{3/2} > \frac{27}{8} = \left(\frac{3}{2}\right)^3$, so $\sqrt{x} > \frac{3}{2}$ and $x > \frac{9}{4}$.

5

Find the tangent gradientAt $P(9, 40)$, $\frac{dy}{dx} = \frac{4}{3}(9) - \frac{9}{2 \cdot 3} = 12 - \frac{3}{2} = \frac{21}{2}$.

6

Write the tangent line $y - 40 = \frac{21}{2}(x - 9)$, so $y = \frac{21}{2}x - \frac{109}{2}$.

7

Find where the tangent meets the x-axisSet $y = 0$: $\frac{21}{2}x - \frac{109}{2} = 0$, so $x = \frac{109}{21}$.

8

Split the areaFrom $x = 0$ to $x = \frac{109}{21}$, the lower boundary is the x -axis. From $x = \frac{109}{21}$ to $x = 9$, the lower boundary is the tangent.

9

Set up the integral

$$R = \int_0^{109/21} \left(\frac{2}{3}x^2 - 9\sqrt{x} + 13 \right) dx + \int_{109/21}^9 \left(\frac{2}{3}x^2 - 9\sqrt{x} + 13 - \left(\frac{21}{2}x - \frac{109}{2} \right) \right) dx.$$

10

Use antiderivatives

$$\int \left(\frac{2}{3}x^2 - 9\sqrt{x} + 13 \right) dx = \frac{2}{9}x^3 - 6x^{3/2} + 13x, \text{ and } \int \left(\frac{21}{2}x - \frac{109}{2} \right) dx = \frac{21}{4}x^2 - \frac{109}{2}x.$$

11

Evaluate

Substituting the limits and simplifying gives $R = \frac{857}{21}$.

FINAL ANSWER

WMA12-01_2023_Oct_Q09

(a) y is increasing for $x > \frac{9}{4}$. (b) **Area** = $\frac{857}{21}$.

DR ESLAM AHMED

TOPIC 11 | FAMILY 02

Optimisation

WHAT THIS FAMILY ASKS

Recognise the optimisation demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 Use constraints to remove extra variables first.

V02 State the maximum/minimum value in context, not just x .

4 classified questions

40 total marks

T11

11_OPTIM

11_OPTIM_HV01

CHAPTER
Applications of Differentiation

FAMILY
Optimisation

VARIANT
Use constraints to remove
extra variables first.

10 marks

WMA12/01 MAY/JUNE 2021

Question 9

Past-paper question

WMA12-01_2021_MayJune_Q09

9.

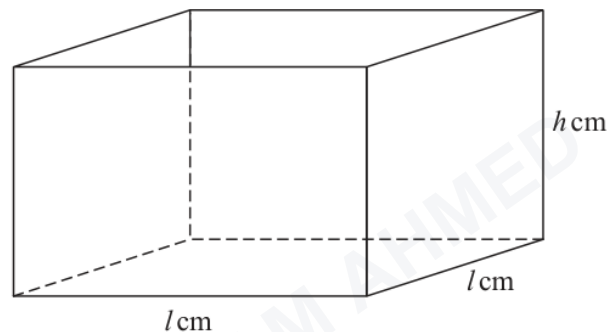


Figure 3

Figure 3 shows a sketch of a square based, open top box.

The height of the box is h cm, and the base edges each have length l cm.

Given that the volume of the box is $250\,000\text{ cm}^3$

(a) show that the external surface area, $S\text{ cm}^2$, of the box is given by

$$S = \frac{250\,000}{h} + 2000\sqrt{h} \quad (3)$$

(b) Use algebraic differentiation to show that S has a stationary point when $h = 250^k$ where k is a rational constant to be found. (5)

(c) Justify by further differentiation that this value of h gives the minimum external surface area of the box. (2)

(Total 10 marks)

**T11 Applications of
Differentiation**

11_OPTIM Optimisation

**11_OPTIM_HV01 Use
constraints to remove extra
variables first.**

WORKED SOLUTION

Question 9

WMA12/01 May/June 2021

WMA12-01_2021_MayJune_Q09

1

Use the volume

The volume is $l^2h = 250000$, so $l^2 = \frac{250000}{h}$.

2

Write the surface area

The open box has one square base and four side faces, so $S = l^2 + 4lh$.

3

Substitute l

Since $l = \sqrt{\frac{250000}{h}} = \frac{500}{\sqrt{h}}$, $4lh = 4 \cdot \frac{500}{\sqrt{h}} \cdot h = 2000\sqrt{h}$.

4

Show S

Therefore $S = \frac{250000}{h} + 2000\sqrt{h}$.

5

Differentiate S

$S = 250000h^{-1} + 2000h^{1/2}$, so $\frac{dS}{dh} = -250000h^{-2} + 1000h^{-1/2}$.

6

Set the derivative to zero

$-250000h^{-2} + 1000h^{-1/2} = 0$, so $1000h^{-1/2} = 250000h^{-2}$.

7

Solve for h

Multiplying by h^2 gives $1000h^{3/2} = 250000$, so $h^{3/2} = 250$ and $h = 250^{2/3}$.

8

Find k

Since $h = 250^k$, we have $k = \frac{2}{3}$.

9

Use the second derivative

$\frac{d^2S}{dh^2} = 500000h^{-3} - 500h^{-3/2}$.

10

Check the sign

At $h = 250^{2/3}$, $h^{3/2} = 250$, so the second derivative is $\frac{500000}{250^2} - \frac{500}{250} = 8 - 2 = 6 > 0$.

11

Conclude

The stationary point is therefore a minimum.

FINAL ANSWER

WMA12-01_2021_MayJune_Q09

(a) **Proven:** $S = \frac{250000}{h} + 2000\sqrt{h}$. (b) $h = 250^{2/3}$, so $k = \frac{2}{3}$. (c) **Minimum.**

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T11

11_OPTIM

11_OPTIM_HV01

CHAPTER
Applications of Differentiation

FAMILY
Optimisation

VARIANT
Use constraints to remove
extra variables first.

10 marks

WMA12/01 JANUARY 2025
Question 3

Past-paper question

WMA12-01_2025_Jan_Q03

3.

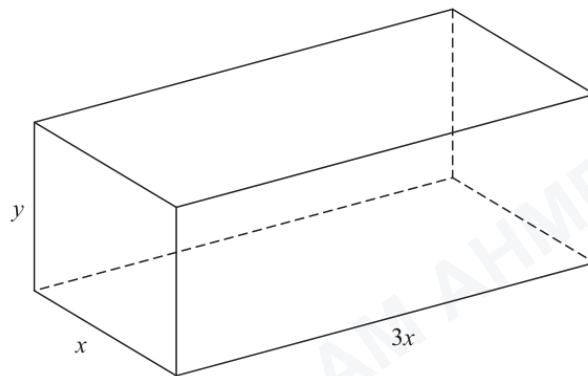


Figure 1

Figure 1 shows an open-topped container used for holding water.

The container is in the shape of a cuboid and is made of sheet metal.

The base of the container is a rectangle $3x$ metres by x metres.

The height of the container is y metres as shown in Figure 1.

Given that the capacity of the container is 120 m^3

(a) show that the area $A \text{ m}^2$ of the sheet metal used to make the container is given by

$$A = Px^2 + \frac{Q}{x}$$

where P and Q are positive constants to be found.

(4)

(b) Use calculus to find the value of x for which A has a stationary value, giving your answer to 3 significant figures.

(4)

(c) Find $\frac{d^2A}{dx^2}$ and hence show that the value of x found in part (b) gives the minimum value of A .

(2)

T11 Applications of
Differentiation

11_OPTIM Optimisation

11_OPTIM_HV01 Use
constraints to remove extra
variables first.

WORKED SOLUTION

Question 3

WMA12/01 January 2025

WMA12-01_2025_Jan_Q03

1

Use the volume

The volume is $(3x)(x)(y) = 3x^2y$. Since the capacity is 120, $3x^2y = 120$, so $y = \frac{40}{x^2}$.

2

Write the metal area

The open top means the metal area is the base plus four sides: $A = 3x^2 + 2(3xy) + 2(xy)$.

3

Substitute for y

$A = 3x^2 + 8xy = 3x^2 + 8x\left(\frac{40}{x^2}\right) = 3x^2 + \frac{320}{x}$. Hence $P = 3$ and $Q = 320$.

4

Differentiate

$$\frac{dA}{dx} = 6x - 320x^{-2} = 6x - \frac{320}{x^2}.$$

5

Use the stationary condition

Set $\frac{dA}{dx} = 0$: $6x = \frac{320}{x^2}$, so $6x^3 = 320$.

6

Find x

$x^3 = \frac{160}{3}$, so $x = \sqrt[3]{\frac{160}{3}} = 3.764\dots$. To 3 significant figures, $x = 3.76$.

7

Differentiate again

$$\frac{d^2A}{dx^2} = 6 + 640x^{-3} = 6 + \frac{640}{x^3}.$$

8

Show the minimum

For $x > 0$, $6 + \frac{640}{x^3} > 0$, so the stationary value found in part (b) is a minimum.

FINAL ANSWER

(a) $A = 3x^2 + \frac{320}{x}$, so $P = 3$ and $Q = 320$. (b) $x = 3.76$. (c) $\frac{d^2A}{dx^2} = 6 + \frac{640}{x^3}$, minimum.

WMA12-01_2025_Jan_Q03

T11

11_OPTIM

11_OPTIM_HV03

CHAPTER
Applications of Differentiation

FAMILY
Optimisation

VARIANT
State the maximum/minimum
value in context, not just $/(x/)$.

9 marks

WMA12/01 JANUARY 2023
Question 2

Past-paper question

WMA12-01_2023_Jan_Q02

2. In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

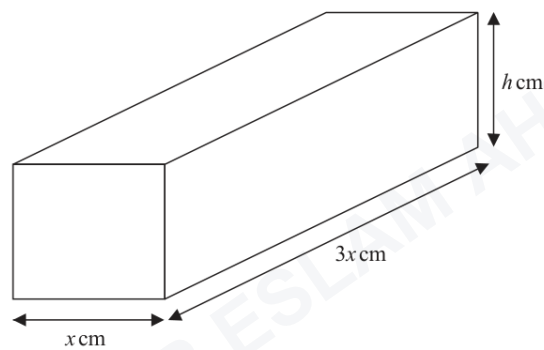


Figure 2

A brick is in the shape of a cuboid with width x cm, length $3x$ cm and height h cm, as shown in Figure 2.

The volume of the brick is 972 cm^3

(a) Show that the surface area of the brick, $S \text{ cm}^2$, is given by

$$S = 6x^2 + \frac{2592}{x}$$

(3)

(b) Find $\frac{dS}{dx}$

(1)

(c) Hence find the value of x for which S is stationary.

(2)

(d) Find $\frac{d^2S}{dx^2}$ and hence show that the value of x found in part (c) gives the minimum value of S .

(2)

(e) Hence find the minimum surface area of the brick.

(1)

T11 Applications of
Differentiation

11_OPTIM Optimisation

11_OPTIM_HV03 State the
maximum/minimum value in
context, not just $/(x/)$.

WORKED SOLUTION

Question 2

WMA12/01 January 2023

WMA12-01_2023_Jan_Q02

1

Use the volumeThe cuboid has dimensions x , $3x$ and h , so its volume is $3x^2h = 972$.

2

Write h in terms of x Hence $h = \frac{972}{3x^2} = \frac{324}{x^2}$.

3

Build the surface areaThe surface area is $S = 2(3x^2 + 3xh + xh) = 6x^2 + 8xh$.

4

Substitute h $S = 6x^2 + 8x \left(\frac{324}{x^2} \right) = 6x^2 + \frac{2592}{x}$.

5

Differentiate $S = 6x^2 + 2592x^{-1}$, so $\frac{dS}{dx} = 12x - 2592x^{-2} = 12x - \frac{2592}{x^2}$.

6

Set the derivative to zeroFor a stationary value, $12x - \frac{2592}{x^2} = 0$.

7

Solve for x Multiplying by x^2 gives $12x^3 = 2592$, so $x^3 = 216$ and $x = 6$.

8

Use the second derivative $\frac{d^2S}{dx^2} = 12 + \frac{5184}{x^3}$.

9

Show it is a minimumAt $x = 6$, $\frac{d^2S}{dx^2} = 12 + \frac{5184}{216} = 36 > 0$, so the stationary value is a minimum.

10

Find the minimum surface area $S = 6(6)^2 + \frac{2592}{6} = 216 + 432 = 648$, so the minimum surface area is 648 cm^2 .

FINAL ANSWER

WMA12-01_2023_Jan_Q02

- (a) $S = 6x^2 + \frac{2592}{x}$. (b) $\frac{dS}{dx} = 12x - \frac{2592}{x^2}$. (c) $x = 6$. (d) **Minimum shown.**
(e) **648 cm².**

DR ESLAM AHMED

T11

11_OPTIM

11_OPTIM_HV03

CHAPTER
Applications of Differentiation

FAMILY
Optimisation

VARIANT
State the maximum/minimum
value in context, not just $/(x/)$.

11 marks

WMA12/01 MAY/JUNE 2025
Question 9

Past-paper question

WMA12-01_2025_MayJune_Q09

9:

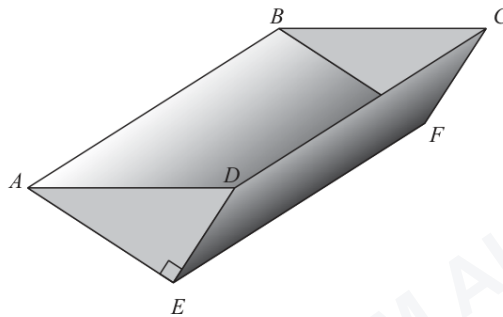


Figure 2

Figure 2 shows a sketch of an open container.

The sides $CDEF$ and $ABFE$ are rectangles.

The ends ADE and BCF are congruent (identical) right-angled triangles.

The container is made from metal of negligible thickness.

Given that

- $AE = BF = 3x$ metres
- $DE = CF = 2x$ metres
- $AB = DC = EF = L$ metres

and the capacity of the container is 12 m^3

(a) show that the area of metal used to make the container, $S \text{ m}^2$, is given by

$$S = Px^2 + \frac{Q}{x}$$

where P and Q are positive integers to be found.

(4)

Given that x can vary,

(b) use algebraic calculus to find the minimum value of S , giving your answer to one decimal place.

(5)

(c) Justify that the value of S found in part (b) is a minimum.

(2)

T11 Applications of
Differentiation

11_OPTIM Optimisation

11_OPTIM_HV03 State the
maximum/minimum value in
context, not just $f(x)$.

WORKED SOLUTION

Question 9

WMA12/01 May/June 2025

WMA12-01_2025_MayJune_Q09

1

Find the triangular end area

Each end is a right-angled triangle with perpendicular sides $3x$ and $2x$, so its area is $\frac{1}{2}(3x)(2x) = 3x^2$.

2

Use the capacity

The volume is cross-section area times length, so $3x^2L = 12$ and $L = \frac{4}{x^2}$.

3

Write the metal area

There are two triangular ends with total area $6x^2$, and two rectangular sides with total area $3xL + 2xL = 5xL$.

4

Substitute for L

$S = 6x^2 + 5x\left(\frac{4}{x^2}\right) = 6x^2 + \frac{20}{x}$. Hence $P = 6$ and $Q = 20$.

5

Differentiate

$$\frac{dS}{dx} = 12x - 20x^{-2} = 12x - \frac{20}{x^2}.$$

6

Find the stationary value of x

Set $\frac{dS}{dx} = 0$: $12x = \frac{20}{x^2}$, so $x^3 = \frac{5}{3}$.

7

Evaluate S

At $x = \sqrt[3]{\frac{5}{3}}$, $S = 6x^2 + \frac{20}{x} = 25.3029\dots$, so the minimum value is 25.3 to one decimal place.

8

Differentiate again

$$\frac{d^2S}{dx^2} = 12 + 40x^{-3} = 12 + \frac{40}{x^3}.$$

9

Justify the minimum

Since $x > 0$, $12 + \frac{40}{x^3} > 0$, so the stationary value found is a minimum.

FINAL ANSWER

WMA12-01_2025_MayJune_Q09

(a) $S = 6x^2 + \frac{20}{x}$, so $P = 6$ and $Q = 20$. (b) 25.3. (c) **Minimum justified.**

DR ESLAM AHMED

TOPIC 11 | FAMILY 03

Curve Behaviour

WHAT THIS FAMILY ASKS

Recognise the curve behaviour demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

3 classified questions

25 total marks

T11

11_SKETCH

11_SKETCH_CORE

CHAPTER

Applications of Differentiation

FAMILY

Curve behaviour

VARIANT

core family move

6 marks

WMA12/01 OCTOBER 2019

Question 1

Past-paper question

WMA12-01_2019_Oct_Q01

1. A curve C has equation $y = 2x^2(x - 5)$

(a) Find, using calculus, the x coordinates of the stationary points of C .

(4)

(b) Hence find the values of x for which y is increasing.

(2)

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T11 Applications of
Differentiation

11_SKETCH Curve behaviour

11_SKETCH_CORE core
family move

WORKED SOLUTION

Question 1

WMA12/01 October 2019

WMA12-01_2019_Oct_Q01

1

Expand the curve

$$y = 2x^2(x - 5) = 2x^3 - 10x^2.$$

2

Differentiate

$$\frac{dy}{dx} = 6x^2 - 20x = 2x(3x - 10).$$

3

Find the stationary values of x

At a stationary point, $\frac{dy}{dx} = 0$. Hence $2x(3x - 10) = 0$, so $x = 0$ or $x = \frac{10}{3}$.

4

Use the sign of the derivative

Since $2x(3x - 10)$ is positive outside its two roots and negative between them, y is increasing for $x < 0$ and for $x > \frac{10}{3}$.

FINAL ANSWER

WMA12-01_2019_Oct_Q01

(a) $x = 0, \frac{10}{3}$. (b) $x < 0$ or $x > \frac{10}{3}$.

T11

11_SKETCH

11_SKETCH_CORE

CHAPTER

Applications of Differentiation

FAMILY

Curve behaviour

VARIANT

core family move

10 marks

WMA12/01 OCTOBER 2021

Question 8

Past-paper question

WMA12-01_2021_Oct_Q08

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

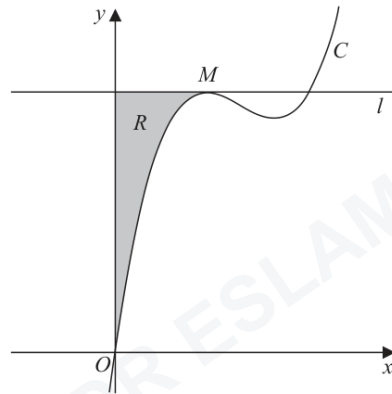


Figure 2

Figure 2 shows a sketch of part of the curve C with equation

$$y = \frac{4}{3}x^3 - 11x^2 + kx \quad \text{where } k \text{ is a constant}$$

The point M is the maximum turning point of C and is shown in Figure 2.

Given that the x coordinate of M is 2

- (a) show that $k = 28$ (3)

- (b) Determine the range of values of x for which y is increasing. (2)

The line l passes through M and is parallel to the x -axis.

The region R , shown shaded in Figure 2, is bounded by the curve C , the line l and the y -axis.

- (c) Find, by algebraic integration, the exact area of R . (5)

T11 Applications of
Differentiation

11_SKETCH Curve behaviour

11_SKETCH_CORE core
family move

WORKED SOLUTION

Question 8

WMA12/01 October 2021

WMA12-01_2021_Oct_Q08

1

Differentiate the curveFor $y = \frac{4}{3}x^3 - 11x^2 + kx$, $\frac{dy}{dx} = 4x^2 - 22x + k$.

2

Use the turning pointAt the turning point M , $x = 2$ and $\frac{dy}{dx} = 0$.

3

Show k $4(2)^2 - 22(2) + k = 0$, so $16 - 44 + k = 0$ and $k = 28$.

4

Factor the derivativeWith $k = 28$, $\frac{dy}{dx} = 4x^2 - 22x + 28 = 2(2x - 7)(x - 2)$.

5

Find where y is increasingThe derivative is positive outside its roots, so y is increasing for $x < 2$ or $x > \frac{7}{2}$.

6

Find the height of the lineAt $x = 2$, $y = \frac{4}{3}(8) - 11(4) + 28(2) = \frac{68}{3}$. Hence line l has equation $y = \frac{68}{3}$.

7

Set up the shaded areaThe region is between the line and the curve from $x = 0$ to $x = 2$, so $A = \int_0^2 \left(\frac{68}{3} - \left(\frac{4}{3}x^3 - 11x^2 + 28x \right) \right) dx$.

8

Integrate

$$A = \left[\frac{68}{3}x - \frac{1}{3}x^4 + \frac{11}{3}x^3 - 14x^2 \right]_0^2.$$

9

Evaluate exactlySubstituting $x = 2$ gives $\frac{136}{3} - \frac{16}{3} + \frac{88}{3} - 56 = \frac{40}{3}$.**FINAL ANSWER**

WMA12-01_2021_Oct_Q08

(a) **Proven:** $k = 28$. (b) $x < 2$ or $x > \frac{7}{2}$. (c) $\frac{40}{3}$.

T11

11_SKETCH

11_SKETCH_CORE

CHAPTER

Applications of Differentiation

FAMILY

Curve behaviour

VARIANT

core family move

9 marks

WMA12/01 MAY/JUNE (R) 2024

Question 6

Past-paper question

WMA12-01_2024_MayJuneR_Q06

6.

$$f(x) = 4x^3 + px^2 + 8x + q$$

where p and q are constants.

Given that

- $(2x + 3)$ is a factor of $f(x)$
- $f(x)$ has a remainder of -5 when divided by $(x + 2)$

(a) (i) show that $p = 10$

(ii) find the value of q .

(5)

(b) Hence find the range of values of x for which $f(x)$ is decreasing.

(Solutions based entirely on calculator technology are not acceptable.)

(4)

T11 Applications of
Differentiation

11_SKETCH Curve behaviour

11_SKETCH_CORE core
family move

WORKED SOLUTION

Question 6

WMA12/01 May/June (R) 2024

WMA12-01_2024_MayJuneR_Q06

1

Use the factor theoremSince $(2x + 3)$ is a factor, $x = -\frac{3}{2}$ is a root.

2

Substitute x equals negative three halves

$$4\left(-\frac{3}{2}\right)^3 + p\left(-\frac{3}{2}\right)^2 + 8\left(-\frac{3}{2}\right) + q = 0.$$

3

Simplify the first condition

$$-\frac{27}{2} + \frac{9}{4}p - 12 + q = 0, \text{ so } \frac{9}{4}p + q = \frac{51}{2}.$$

4

Use the remainder theoremThe remainder on division by $(x + 2)$ is $f(-2)$.

5

Substitute x equals negative two

$$f(-2) = 4(-8) + 4p - 16 + q = 4p + q - 48.$$

6

Use the given remainder

$$4p + q - 48 = -5, \text{ so } 4p + q = 43.$$

7

Solve for p and q

Subtract $\frac{9}{4}p + q = \frac{51}{2}$ from $4p + q = 43$: $\frac{7}{4}p = \frac{35}{2}$, so $p = 10$. Then $4(10) + q = 43$, so $q = 3$.

8

Differentiate

With $p = 10$, $f(x) = 4x^3 + 10x^2 + 8x + 3$, so $f'(x) = 12x^2 + 20x + 8$.

9

Factor the derivative

$$f'(x) = 4(3x + 2)(x + 1).$$

10

Find where decreasing

The quadratic derivative is negative between its roots, so $f(x)$ is decreasing for $-1 < x < -\frac{2}{3}$.

FINAL ANSWER

WMA12-01_2024_MayJuneR_Q06

$$(a)(i) p = 10. \quad (a)(ii) q = 3. \quad (b) -1 < x < -\frac{2}{3}.$$

DR ESLAM AHMED

TOPIC 11 | FAMILY 04

Stationary Points

WHAT THIS FAMILY ASKS

Recognise the stationary points demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**11 classified questions**

91 total marks

T11

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CHAPTER

Applications of Differentiation

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10 marks

WMA12/01 JANUARY 2020

Question 10

Past-paper question

WMA12-01_2020_Jan_Q10

10. A curve C has equation

$$y = 4x^3 - 9x + \frac{k}{x} \quad x > 0$$

where k is a constant.

The point P with x coordinate $\frac{1}{2}$ lies on C .

Given that P is a stationary point of C ,

(a) show that $k = -\frac{3}{2}$ (4)

(b) Determine the nature of the stationary point at P , justifying your answer. (2)

The curve C has a second stationary point.

(c) Using algebra, find the x coordinate of this second stationary point. (4)

(Total 10 marks)

T11 Applications of
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WORKED SOLUTION

Question 10

WMA12/01 January 2020

WMA12-01_2020_Jan_Q10

1

Differentiate the curveWrite $y = 4x^3 - 9x + kx^{-1}$. Then $\frac{dy}{dx} = 12x^2 - 9 - kx^{-2}$.

2

Use the stationary pointAt $x = \frac{1}{2}$, $\frac{dy}{dx} = 0$, so $0 = 12\left(\frac{1}{2}\right)^2 - 9 - k\left(\frac{1}{2}\right)^{-2} = 3 - 9 - 4k$.

3

Find k $-6 - 4k = 0$, hence $k = -\frac{3}{2}$.

4

Use the second derivativeWith $k = -\frac{3}{2}$, $\frac{dy}{dx} = 12x^2 - 9 + \frac{3}{2x^2}$ and $\frac{d^2y}{dx^2} = 24x - 3x^{-3}$.

5

Determine the nature of PAt $x = \frac{1}{2}$, $\frac{d^2y}{dx^2} = 12 - 24 = -12 < 0$, so P is a local maximum.

6

Find the other stationary pointSet $12x^2 - 9 + \frac{3}{2x^2} = 0$. Multiplying by $2x^2$ gives $24x^4 - 18x^2 + 3 = 0$, or $8x^4 - 6x^2 + 1 = 0$.

7

Solve the quadratic in x squared $(4x^2 - 1)(2x^2 - 1) = 0$. The given point has $x = \frac{1}{2}$, so the second stationary point has $x^2 = \frac{1}{2}$.

8

State the positive x coordinateSince $x > 0$, the second stationary point has $x = \frac{1}{\sqrt{2}}$.**FINAL ANSWER**

WMA12-01_2020_Jan_Q10

(a) $k = -\frac{3}{2}$. (b) local maximum. (c) $x = \frac{1}{\sqrt{2}}$.

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7 marks

WMA12/01 JANUARY 2021

Question 2

Past-paper question

WMA12-01_2021_Jan_Q02

2. A curve has equation

$$y = x^3 - x^2 - 16x + 2$$

- (a) Using calculus, find the x coordinates of the stationary points of the curve. (4)
- (b) Justify, by further calculus, the nature of all of the stationary points of the curve. (3)

T11 Applications of
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WORKED SOLUTION

Question 2

WMA12/01 January 2021

WMA12-01_2021_Jan_Q02

1

Differentiate the curveFor $y = x^3 - x^2 - 16x + 2$, $\frac{dy}{dx} = 3x^2 - 2x - 16$.

2

Set the gradient to zeroStationary points occur when $3x^2 - 2x - 16 = 0$.

3

Factorise $3x^2 - 2x - 16 = (3x - 8)(x + 2)$, so $x = \frac{8}{3}$ or $x = -2$.

4

Differentiate again $\frac{d^2y}{dx^2} = 6x - 2$.

5

Classify x equals negative twoAt $x = -2$, $6x - 2 = -14 < 0$, so the stationary point is a local maximum.

6

Classify x equals eight over threeAt $x = \frac{8}{3}$, $6x - 2 = 14 > 0$, so the stationary point is a local minimum.**FINAL ANSWER**

WMA12-01_2021_Jan_Q02

(a) $x = -2$, $\frac{8}{3}$. (b) local maximum at $x = -2$ and local minimum at $x = \frac{8}{3}$.

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8 marks

WMA12/01 JANUARY 2022

Question 2

Past-paper question

WMA12-01_2022_Jan_Q02

2. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = 27x^{\frac{1}{2}} - x^{\frac{3}{2}} - 20 \quad x > 0$$

(a) Find $\frac{dy}{dx}$, giving each term in simplest form.

(2)

(b) Hence find the coordinates of the stationary point of C .

(4)

(c) Find $\frac{d^2y}{dx^2}$ and hence determine the nature of the stationary point of C .

(2)

T11 Applications of
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WORKED SOLUTION

Question 2

WMA12/01 January 2022

WMA12-01_2022_Jan_Q02

1

DifferentiateFor $y = 27x^{1/2} - x^{3/2} - 20$, $\frac{dy}{dx} = \frac{27}{2}x^{-1/2} - \frac{3}{2}x^{1/2}$.

2

Set the gradient to zero

$$\frac{27}{2}x^{-1/2} - \frac{3}{2}x^{1/2} = 0.$$

3

Clear the powersMultiplying by $2x^{1/2}$ gives $27 - 3x = 0$, so $x = 9$.

4

Find yAt $x = 9$, $y = 27\sqrt{9} - 9^{3/2} - 20 = 81 - 27 - 20 = 34$.

5

State the stationary pointThe stationary point is $(9, 34)$.

6

Differentiate again

$$\frac{d^2y}{dx^2} = -\frac{27}{4}x^{-3/2} - \frac{3}{4}x^{-1/2}.$$

7

Determine the natureAt $x = 9$, both terms in $\frac{d^2y}{dx^2}$ are negative, so the stationary point is a local maximum.**FINAL ANSWER**

WMA12-01_2022_Jan_Q02

(a) $\frac{dy}{dx} = \frac{27}{2}x^{-1/2} - \frac{3}{2}x^{1/2}$. (b) $(9, 34)$. (c) **local maximum.**

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WMA12/01 MAY/JUNE 2022

Question 8

Past-paper question

WMA12-01_2022_MayJune_Q08

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A curve has equation

$$y = 256x^4 - 304x - 35 + \frac{27}{x^2} \quad x \neq 0$$

- (a) Find $\frac{dy}{dx}$ (3)
- (b) Hence find the coordinates of the stationary points of the curve. (5)

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WORKED SOLUTION

Question 8

WMA12/01 May/June 2022

WMA12-01_2022_MayJune_Q08

1

Rewrite with powers

$$y = 256x^4 - 304x - 35 + 27x^{-2}.$$

2

Differentiate

$$\frac{dy}{dx} = 1024x^3 - 304 - 54x^{-3} = 1024x^3 - 304 - \frac{54}{x^3}.$$

3

Set the derivative to zero

$$1024x^3 - 304 - \frac{54}{x^3} = 0.$$

4

Use a substitution

Multiplying by x^3 gives $1024x^6 - 304x^3 - 54 = 0$. Let $u = x^3$.

5

Solve for u

$$1024u^2 - 304u - 54 = 0, \text{ or } 512u^2 - 152u - 27 = 0 = (8u + 1)(64u - 27).$$

6

Find x

Thus $x^3 = -\frac{1}{8}$ or $x^3 = \frac{27}{64}$, giving $x = -\frac{1}{2}$ or $x = \frac{3}{4}$.

7

Find the y values

Substitution into $y = 256x^4 - 304x - 35 + \frac{27}{x^2}$ gives $y = 241$ at $x = -\frac{1}{2}$ and $y = -134$ at $x = \frac{3}{4}$.

8

State the stationary points

The stationary points are $(-\frac{1}{2}, 241)$ and $(\frac{3}{4}, -134)$.

FINAL ANSWER

WMA12-01_2022_MayJune_Q08

$$(a) \frac{dy}{dx} = 1024x^3 - 304 - \frac{54}{x^3}. \quad (b) \left(-\frac{1}{2}, 241\right), \left(\frac{3}{4}, -134\right).$$

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9 marks

WMA12/01 OCTOBER 2022

Question 7

Past-paper question

WMA12-01_2022_Oct_Q07

7. The curve C has equation

$$y = \frac{12x^3(x-7) + 14x(13x-15)}{21\sqrt{x}} \quad x > 0$$

(a) Write the equation of C in the form

$$y = ax^{\frac{7}{2}} + bx^{\frac{5}{2}} + cx^{\frac{3}{2}} + dx^{\frac{1}{2}}$$

where a , b , c and d are fully simplified constants.

(3)

The curve C has three turning points.

Using calculus,

(b) show that the x coordinates of the three turning points satisfy the equation

$$2x^3 - 10x^2 + 13x - 5 = 0$$

(3)

Given that the x coordinate of one of the turning points is 1

(c) find, using algebra, the exact x coordinates of the other two turning points.

(Solutions based entirely on calculator technology are not acceptable.)

(3)

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WORKED SOLUTION

Question 7

WMA12/01 October 2022

WMA12-01_2022_Oct_Q07

1

Expand the numerator

$$12x^3(x-7) + 14x(13x-15) = 12x^4 - 84x^3 + 182x^2 - 210x.$$

2

Divide by 21 root x

Since $\sqrt{x} = x^{1/2}$, divide each term by $21x^{1/2}$.

3

Write in the required form

$$y = \frac{12}{21}x^{7/2} - \frac{84}{21}x^{5/2} + \frac{182}{21}x^{3/2} - \frac{210}{21}x^{1/2} = \frac{4}{7}x^{7/2} - 4x^{5/2} + \frac{26}{3}x^{3/2} - 10x^{1/2}.$$

4

Differentiate

$$\frac{dy}{dx} = 2x^{5/2} - 10x^{3/2} + 13x^{1/2} - 5x^{-1/2}.$$

5

Factor the derivative

$$\frac{dy}{dx} = x^{-1/2}(2x^3 - 10x^2 + 13x - 5).$$

6

Use the turning point condition

At a turning point, $\frac{dy}{dx} = 0$. Since $x > 0$, $x^{-1/2} \neq 0$, so the x coordinates satisfy $2x^3 - 10x^2 + 13x - 5 = 0$.

7

Use the known root

One root is $x = 1$, so divide $2x^3 - 10x^2 + 13x - 5$ by $(x - 1)$.

8

Factor the cubic

$$2x^3 - 10x^2 + 13x - 5 = (x - 1)(2x^2 - 8x + 5).$$

9

Solve the quadratic

$$2x^2 - 8x + 5 = 0 \text{ gives } x = \frac{8 \pm \sqrt{64 - 40}}{4} = \frac{8 \pm 2\sqrt{6}}{4} = 2 \pm \frac{\sqrt{6}}{2}.$$

FINAL ANSWER

WMA12-01_2022_Oct_Q07

$$(a) y = \frac{4}{7}x^{7/2} - 4x^{5/2} + \frac{26}{3}x^{3/2} - 10x^{1/2}. \quad (b) \text{ Proven.} \quad (c) x = 2 - \frac{\sqrt{6}}{2} \text{ and } x = 2 + \frac{\sqrt{6}}{2}.$$

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WMA12/01 MAY/JUNE 2023

Question 7

Past-paper question

WMA12-01_2023_MayJune_Q07

7. The height of a river above a fixed point on the riverbed was monitored over a 7-day period.

The height of the river, H metres, t days after monitoring began, was given by

$$H = \frac{\sqrt{t}}{20}(20 + 6t - t^2) + 17 \quad 0 \leq t \leq 7$$

Given that H has a stationary value at $t = \alpha$

- (a) use calculus to show that α satisfies the equation

$$5\alpha^2 - 18\alpha - 20 = 0 \quad (5)$$

- (b) Hence find the value of α , giving your answer to 3 decimal places. (1)

- (c) Use further calculus to prove that H is a maximum at this value of α . (2)

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WORKED SOLUTION

Question 7

WMA12/01 May/June 2023

WMA12-01_2023_MayJune_Q07

1

Rewrite H with powers

$$H = \frac{1}{20}t^{1/2}(20 + 6t - t^2) + 17.$$

2

Expand before differentiating

$$H = \frac{1}{20}(20t^{1/2} + 6t^{3/2} - t^{5/2}) + 17.$$

3

Differentiate

$$\frac{dH}{dt} = \frac{1}{20} \left(10t^{-1/2} + 9t^{1/2} - \frac{5}{2}t^{3/2} \right).$$

4

Put over a common factor

$$\frac{dH}{dt} = \frac{-5t^2 + 18t + 20}{40\sqrt{t}}.$$

5

Use the stationary condition

At $t = \alpha$, $\frac{dH}{dt} = 0$. Since $\alpha > 0$, the denominator is not zero, so $-5\alpha^2 + 18\alpha + 20 = 0$.

6

Show the required equation

Multiplying by -1 gives $5\alpha^2 - 18\alpha - 20 = 0$.

7

Find alpha

$$\alpha = \frac{18 \pm \sqrt{18^2 + 4(5)(20)}}{10} = \frac{9 \pm \sqrt{181}}{5}.$$

8

Use the time interval

Only the positive value lies in $0 \leq t \leq 7$, so $\alpha = \frac{9 + \sqrt{181}}{5} = 4.491$ to 3 decimal places.

9

Differentiate again

$$\frac{d^2H}{dt^2} = \frac{-15t^2 + 18t - 20}{80t^{3/2}}.$$

10

Use the stationary equation

At $t = \alpha$, $5\alpha^2 = 18\alpha + 20$. Therefore the numerator of $\frac{d^2H}{dt^2}$ is $-15\alpha^2 + 18\alpha - 20 = -3(18\alpha + 20) + 18\alpha - 20 = -36\alpha - 80$.

11

Show it is negative

Since $\alpha > 0$, $-36\alpha - 80 < 0$, and the denominator $80\alpha^{3/2}$ is positive. Thus $\frac{d^2H}{dt^2} < 0$, so H is a maximum.

FINAL ANSWER

WMA12-01_2023_MayJune_Q07

(a) **Proven.** (b) $\alpha = 4.491$. (c) H is a maximum at this value.

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WMA12/01 JANUARY 2024

Question 10

Past-paper question

WMA12-01_2024_Jan_Q10

10.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

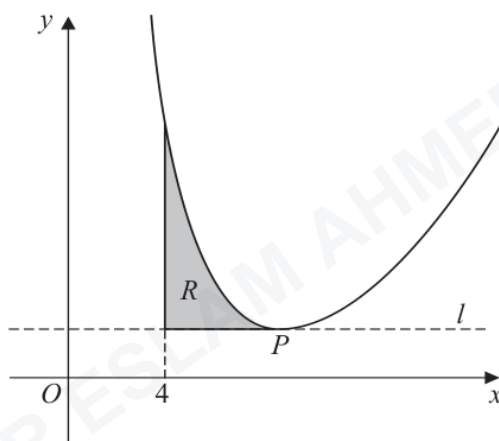


Figure 2

Figure 2 shows a sketch of the curve with equation

$$y = \frac{1}{2}x^2 + \frac{1458}{\sqrt{x^3}} - 74 \quad x > 0$$

The point P is the only stationary point on the curve.(a) Use calculus to show that the x coordinate of P is 9

(4)

The line l passes through the point P and is parallel to the x -axis.The region R , shown shaded in Figure 2, is bounded by the curve, the line l and the line with equation $x = 4$ (b) Use algebraic integration to find the exact area of R .

(5)

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WORKED SOLUTION

Question 10

WMA12/01 January 2024

WMA12-01_2024_Jan_Q10

1

Rewrite the curve

$$y = \frac{1}{2}x^2 + 1458x^{-3/2} - 74.$$

2

Differentiate

$$\frac{dy}{dx} = x + 1458 \left(-\frac{3}{2}\right) x^{-5/2} = x - 2187x^{-5/2}.$$

3

Use the stationary condition

$$\text{At a stationary point, } x - 2187x^{-5/2} = 0.$$

4

Solve for x

$$x = 2187x^{-5/2}, \text{ so } x^{7/2} = 2187 = 3^7.$$

5

Show x coordinate

Raising both sides to the power $\frac{2}{7}$ gives $x = 3^2 = 9$. Since $x > 0$, this is the only stationary point.

6

Find the horizontal line

$$\text{At } x = 9, y = \frac{1}{2}(9)^2 + \frac{1458}{\sqrt{9^3}} - 74 = \frac{81}{2} + 54 - 74 = \frac{41}{2}.$$

7

Set up the area

The region lies from $x = 4$ to $x = 9$ between the curve and the horizontal line $y = \frac{41}{2}$.

8

Integrate the difference

$$R = \int_4^9 \left(\frac{1}{2}x^2 + 1458x^{-3/2} - 74 - \frac{41}{2} \right) dx.$$

9

Find an antiderivative

$$\int \left(\frac{1}{2}x^2 + 1458x^{-3/2} - \frac{189}{2} \right) dx = \frac{x^3}{6} - 2916x^{-1/2} - \frac{189}{2}x.$$

10

Evaluate exactly

$$R = \left[\frac{x^3}{6} - 2916x^{-1/2} - \frac{189}{2}x \right]_4^9 = \frac{373}{3}.$$

FINAL ANSWER

WMA12-01_2024_Jan_Q10

(a) **Proven.** (b) $\frac{373}{3}$.

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8 marks

WMA12/01 MAY/JUNE 2024

Question 9

Past-paper question

WMA12-01_2024_MayJune_Q09

9.

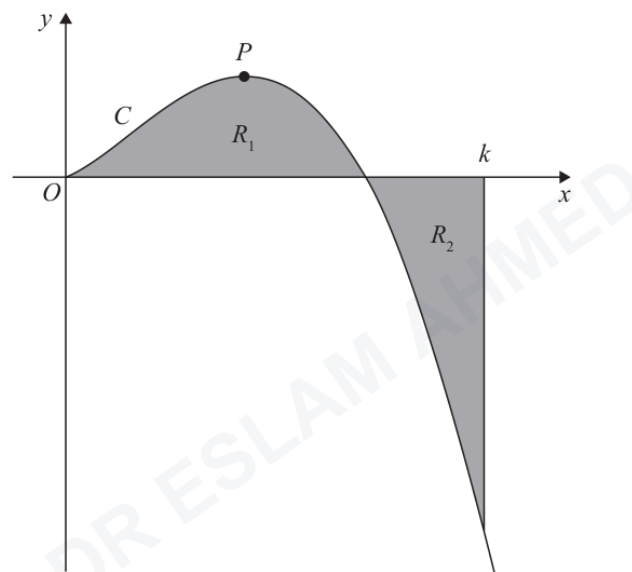


Figure 1

Figure 1 is a sketch of the curve C with equation

$$y = 2x^{\frac{3}{2}}(4 - x) \quad x \geq 0$$

The point P is the stationary point of C .

(a) Find, using calculus, the x coordinate of P .

(4)

The region R_1 , shown shaded in Figure 1, is bounded by C and the x -axis.

The region R_2 , also shown shaded in Figure 1, is bounded by C , the x -axis and the line with equation $x = k$, where k is a constant.

Given that the area of R_1 is equal to the area of R_2

(b) find, using calculus, the exact value of k .

(4)

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WORKED SOLUTION

Question 9

WMA12/01 May/June 2024

WMA12-01_2024_MayJune_Q09

1

Rewrite the curve

$$y = 2x^{3/2}(4 - x) = 8x^{3/2} - 2x^{5/2}.$$

2

Differentiate

$$\frac{dy}{dx} = 12x^{1/2} - 5x^{3/2} = \sqrt{x}(12 - 5x).$$

3

Find the stationary point

For the interior stationary point P , $12 - 5x = 0$, so $x = \frac{12}{5}$.

4

Set up the area condition

R_1 is the area from $x = 0$ to $x = 4$ above the x -axis. R_2 is the area from $x = 4$ to $x = k$ below the x -axis.

5

Use equal areas

Equal areas mean $\int_0^4 y \, dx = -\int_4^k y \, dx$, so $\int_0^k y \, dx = 0$.

6

Integrate y

$$\int 2x^{3/2}(4 - x) \, dx = \int (8x^{3/2} - 2x^{5/2}) \, dx = \frac{16}{5}x^{5/2} - \frac{4}{7}x^{7/2}.$$

7

Solve for k

$$\frac{16}{5}k^{5/2} - \frac{4}{7}k^{7/2} = 0.$$

8

Factor

$$4k^{5/2} \left(\frac{4}{5} - \frac{k}{7} \right) = 0.$$

9

Choose the value beyond four

Since $k > 4$, $\frac{4}{5} - \frac{k}{7} = 0$, so $k = \frac{28}{5}$.

FINAL ANSWER

$$(a) \, x = \frac{12}{5}. \quad (b) \, k = \frac{28}{5}.$$

WMA12-01_2024_MayJune_Q09

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WMA12/01 MAY/JUNE (R) 2024

Question 10

Past-paper question

WMA12-01_2024_MayJuneR_Q10

10.

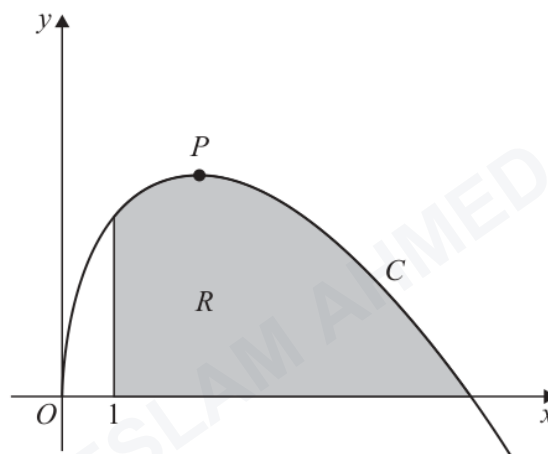


Figure 1

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 1 shows a sketch of part of the curve C with equation

$$y = \frac{9x - x^2}{2\sqrt{x}} \quad x > 0$$

The point P is the stationary point on C .

(a) Find, using calculus, the x coordinate of P .

(4)

The finite region R , shown shaded in Figure 1, is bounded by the curve C , the x -axis and the line with equation $x = 1$

(b) Using calculus, calculate the exact area of R .

(5)

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WORKED SOLUTION

Question 10

WMA12/01 May/June (R) 2024

WMA12-01_2024_MayJuneR_Q10

1

Rewrite the curve

$$y = \frac{9x-x^2}{2\sqrt{x}} = \frac{1}{2}(9x^{1/2} - x^{3/2}).$$

2

Differentiate

$$\frac{dy}{dx} = \frac{1}{2} \left(\frac{9}{2}x^{-1/2} - \frac{3}{2}x^{1/2} \right) = \frac{3(3-x)}{4\sqrt{x}}.$$

3

Use the stationary condition

For $x > 0$, the denominator is not zero. So $\frac{dy}{dx} = 0$ when $3 - x = 0$.

4

Find P

Therefore the x coordinate of P is 3.

5

Find where the curve meets the x-axis

$y = 0$ when $9x - x^2 = 0$, so $x = 9$ for the positive intercept.

6

Set up the area

The region is under the curve from $x = 1$ to $x = 9$, so $R = \int_1^9 \frac{9x-x^2}{2\sqrt{x}} dx$.

7

Integrate

$$R = \int_1^9 \frac{1}{2}(9x^{1/2} - x^{3/2}) dx = \left[3x^{3/2} - \frac{1}{5}x^{5/2} \right]_1^9.$$

8

Evaluate

$$R = \left(3(27) - \frac{1}{5}(243) \right) - \left(3 - \frac{1}{5} \right) = \frac{148}{5}.$$

FINAL ANSWER

$$(a) x = 3. \quad (b) \frac{148}{5}.$$

WMA12-01_2024_MayJuneR_Q10

T11

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CHAPTER

Applications of Differentiation

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7 marks

WMA12/01 OCTOBER 2024

Question 4

Past-paper question

WMA12-01_2024_Oct_Q04

4. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = 4x^{\frac{1}{2}} + 9x^{-\frac{1}{2}} + 3 \quad x > 0$$

- (a) Find $\frac{dy}{dx}$ giving each term in simplest form. (2)
- (b) Hence find the x coordinate of the stationary point of C . (2)
- (c) (i) Find $\frac{d^2y}{dx^2}$ giving each term in simplest form.
- (ii) Hence determine the nature of the stationary point of C , giving a reason for your answer. (2)
- (d) State the range of values of x for which y is decreasing. (1)

T11 Applications of
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WORKED SOLUTION

Question 4

WMA12/01 October 2024

WMA12-01_2024_Oct_Q04

1

Rewrite using powers

$$y = 4x^{1/2} + 9x^{-1/2} + 3.$$

2

Differentiate

$$\frac{dy}{dx} = 2x^{-1/2} - \frac{9}{2}x^{-3/2} = \frac{2}{\sqrt{x}} - \frac{9}{2x^{3/2}}.$$

3

Set the derivative to zero

$$\frac{dy}{dx} = \frac{4x-9}{2x^{3/2}}. \text{ Since } x > 0, \text{ the stationary point occurs when } 4x - 9 = 0.$$

4

Find the x coordinate

$$\text{Therefore } x = \frac{9}{4}.$$

5

Differentiate again

$$\frac{d^2y}{dx^2} = -x^{-3/2} + \frac{27}{4}x^{-5/2} = -\frac{1}{x^{3/2}} + \frac{27}{4x^{5/2}}.$$

6

Test the stationary point

$$\frac{d^2y}{dx^2} = \frac{27-4x}{4x^{5/2}}. \text{ At } x = \frac{9}{4}, \text{ the numerator is } 27 - 9 = 18 > 0, \text{ so the stationary point is a minimum.}$$

7

Find where y is decreasing

$$\text{For } x > 0, 2x^{3/2} > 0, \text{ so } \frac{dy}{dx} < 0 \text{ when } 4x - 9 < 0. \text{ Hence } y \text{ is decreasing for } 0 < x < \frac{9}{4}.$$

FINAL ANSWER

WMA12-01_2024_Oct_Q04

$$(a) \frac{2}{\sqrt{x}} - \frac{9}{2x^{3/2}}. \quad (b) x = \frac{9}{4}. \quad (c)(i) -\frac{1}{x^{3/2}} + \frac{27}{4x^{5/2}}. \quad (c)(ii) \text{ Minimum.} \quad (d) 0 < x < \frac{9}{4}.$$

T11

11_STAT

11_STAT_CORE

CHAPTER
Applications of Differentiation

FAMILY
Stationary points

VARIANT
core family move

8 marks

WMA12/01 JANUARY 2026

Question 10

Past-paper question

WMA12-01_2026_Jan_Q10

10.

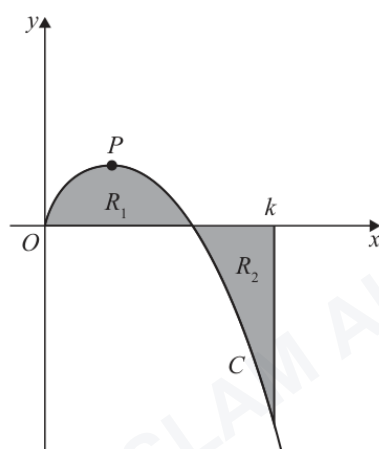


Figure 2

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 2 shows a sketch of the curve C with equation

$$y = \frac{\sqrt{x}(100 - x^2)}{40} \quad x \geq 0$$

The point P is a stationary point on C .

(a) Use algebraic differentiation to find the exact x coordinate of P .

(4)

The region R_1 , shown shaded in Figure 2, is bounded by C and the x -axis.

The region R_2 , also shown shaded in Figure 2, is bounded by C , the x -axis and the line with equation $x = k$, where k is a constant.

Given that the area of R_1 is equal to the area of R_2

(b) use algebraic integration to find the exact value of k .

(4)

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WORKED SOLUTION

Question 10

WMA12/01 January 2026

WMA12-01_2026_Jan_Q10

1

Rewrite the curve

$$y = \frac{\sqrt{x}(100-x^2)}{40} = \frac{5}{2}x^{1/2} - \frac{1}{40}x^{5/2}.$$

2

Differentiate

$$\frac{dy}{dx} = \frac{5}{4}x^{-1/2} - \frac{1}{16}x^{3/2}.$$

3

Use the stationary point

At P , $\frac{dy}{dx} = 0$. Thus $\frac{5}{4}x^{-1/2} = \frac{1}{16}x^{3/2}$. Multiplying by $16x^{1/2}$ gives $20 = x^2$.

4

Find the x-coordinate

Since $x \geq 0$, $x = \sqrt{20} = 2\sqrt{5}$.

5

Integrate the curve

$$\int y \, dx = \int \left(\frac{5}{2}x^{1/2} - \frac{1}{40}x^{5/2} \right) dx = \frac{5}{3}x^{3/2} - \frac{1}{140}x^{7/2}.$$

6

Use equal areas

The signed area from 0 to k is zero because the positive area R_1 equals the negative area R_2 in magnitude.

7

Solve for k

$\frac{5}{3}k^{3/2} - \frac{1}{140}k^{7/2} = 0$. Since $k > 0$, divide by $k^{3/2}$ to get $\frac{5}{3} - \frac{k^2}{140} = 0$, so $k^2 = \frac{700}{3}$.

8

Write the exact value

$$k = \sqrt{\frac{700}{3}} = \frac{10\sqrt{21}}{3}.$$

FINAL ANSWER

WMA12-01_2026_Jan_Q10

$$(a) \, x = 2\sqrt{5}. \quad (b) \, k = \frac{10\sqrt{21}}{3}.$$